PW SKILLS

Introduction to Data Analytics



Session Objectives

- Explain the role of a Data Analyst and the analytics process.
- Identify essential tools: Excel, SQL, Python, Tableau, Power BI.
- Understand how insights drive business decisions.
- Recognize the evolution of analytics (Descriptive → Generative).
- Explore how Generative AI supports analysts.
- Learn the basics of prompt engineering.
- Discover future skills and career pathways for analysts.

Let’s Begin!

Before diving deeper, let’s first understand:

- **What is a Data Analyst?**
- **Why does this role matter in today’s world?**

We’ll also look at **everyday examples** of data analysis around us.



What is Data Analytics ???

- Data analytics is the process of analyzing raw data in order to draw out meaningful, actionable insights, which are then used to inform and drive smart business decisions.
- Helps organizations make evidence-based decisions.



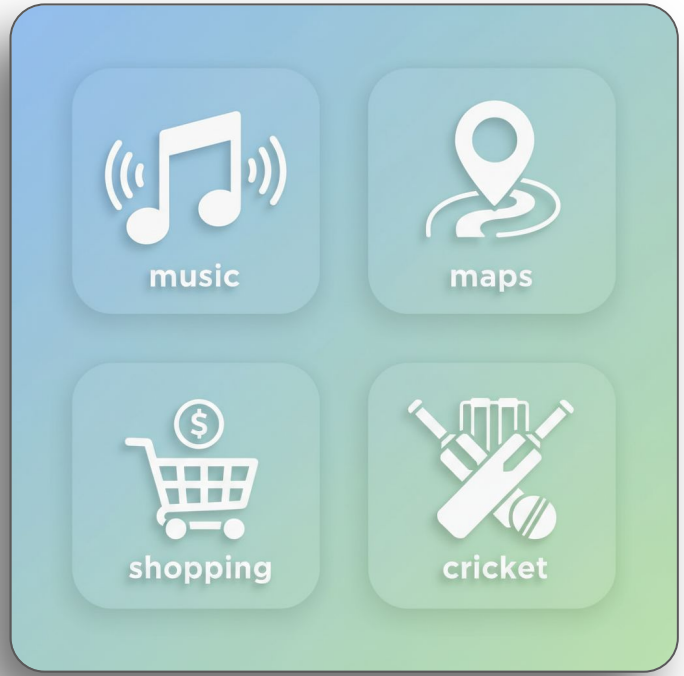
Why Data Analytics Matters

- Raw data is like oil – valuable only when refined.
- Businesses today rely on analytics for almost every decision.
- Examples:
  - Netflix recommends shows.
  - Amazon suggests products.
  - Banks detect fraud.
  - Hospitals predict patient risk.



Data Analytics in Everyday Life

- Spotify suggested playlists based on listening habits.
- Google Maps predicts traffic.
- Flipkart/Amazon recommends items.
- Cricket teams analyze player performance.



Evolution of Analytics



Descriptive & Diagnostic Analytics

- Descriptive analytics summarizes past data to show what happened (e.g., monthly sales report showing total sales).
- Diagnostic analytics explores why the event occurred by identifying causes (e.g., sales dropped because the holiday season ended).
- Example:
    - Sales report shows a decline in June; Diagnostic analysis reveals it is due to seasonal demand ending.

Predictive & Prescriptive Analytics

- Predictive analytics forecasts future outcomes using past data (e.g., predicting next month’s sales based on trends).
- Prescriptive analytics recommends actions to achieve goals (e.g., suggesting discounts to increase sales in a slow month).
- Example:
    - Forecast predicts lower sales in July; prescriptive analysis advises running discount offers to boost sales.

Generative Analytics

- Generative Analytics uses AI to automatically create reports, summaries, and charts.
- AI generates insights based on simple prompts, making data easier to understand.
- **Example:** “Summarize Q2 sales in 3 points” → AI writes key insights like growth and top categories.
- Benefits include saving time, real-time storytelling, and making analytics accessible to everyone.



Why Process Matters ?

- Analytics has multiple steps: Collection → Cleaning → Visualization → Interpretation → Reporting.
- Skipping any step = wrong results.
- Example: If data isn’t cleaned, managers may take wrong business decisions.

Quality >> Quantity

- Having fewer but clean data points is better than huge messy datasets.
- Example: 100 clean customer records > 10,000 messy ones.



The Role of Data Analyst

Now that we know what a Data Analyst is and why it matters,
let's explore:

- Who is a Data Analyst?
- Their key responsibilities
- Essential skills they use every day



Who is a Data Analyst

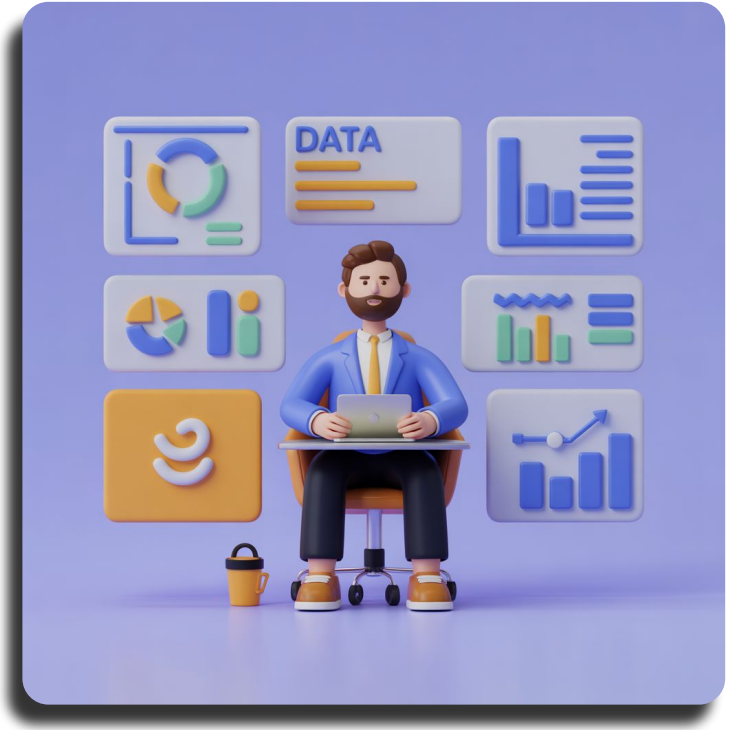
“A data analyst is like a detective of data they gather evidence (data), look for clues (patterns), and tell the story (insights) to help leaders make smart decisions.”

A Data Analyst bridges the gap between raw data and business decisions.



Responsibilities of a Data Analyst

- Collect data from multiple sources.
- Clean and prepare datasets.
- Analyze data to find meaningful patterns.
- Create reports & dashboards.
- Communicate insights clearly.



Skills and Tools Required...

1. **Technical Skills**
   a. **Excel** → Foundation tool for data cleaning, basic analysis, pivot tables, and dashboards. Still widely used in businesses.

   b. **SQL** → Essential for extracting and querying structured data from databases. Every analyst should be comfortable writing queries.

   c. **Python** → For advanced data cleaning, visualization, and machine learning. Libraries like Pandas, NumPy, and Matplotlib are powerful for deeper analysis.

   d. **Power BI/Tableau** → Used for creating interactive dashboards and reports. Helps convert data into visuals that decision-makers can easily understand.

These are the “hard skills” that allow you to actually manipulate and analyze data.



Business Skills..

- **Understanding Problems** → A data analyst should know the business context: *What issue are we solving? Why does it matter?*

- **Asking the Right Questions** → Instead of just crunching numbers, analysts must ask insightful questions that lead to meaningful solutions.
  Example: Instead of “What are the sales numbers?” → Ask “Which products are driving profit, and how can we increase them?”

These skills bridge the gap between raw data and business impact.



Soft Skills...

- **Storytelling** → Numbers alone don't convince. Being able to build a *narrative* around the data helps stakeholders connect with insights.
- **Communication** → Presenting findings clearly to both technical and non-technical audiences.
- **Critical Thinking** → Looking beyond surface trends, questioning assumptions, and finding root causes.

These skills make your analysis impactful and actionable.



Difference in Roles

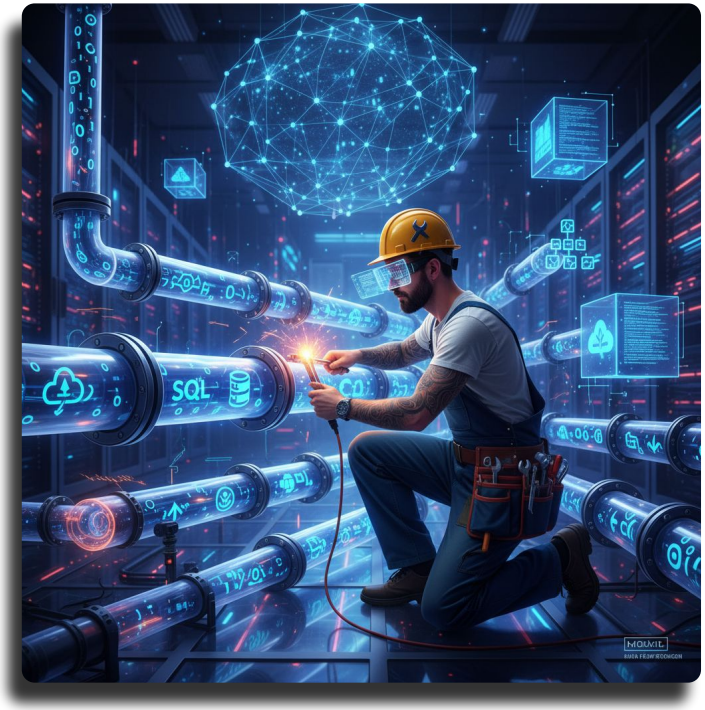
Data Analyst

- Focus: *Understanding the past & present*
- Responsibilities:
  - Cleans and analyzes data
  - Creates reports, dashboards, and visualizations
  - Finds patterns, trends, and insights that help businesses make decisions
- They are the **“interpreters” who translate data into business insights**.



Data Engineer

- Focus: *Infrastructure and data pipelines*
- Responsibilities:
  - Collects and organizes raw data from different sources
  - Manages databases and ensures data quality
  - Builds scalable pipelines so analysts and scientists can access clean, ready-to-use data
- Think of them as the **“builders and plumbers” of the data world**.



Data Scientist

- Focus: *Predicting the future & advanced analytics*
- Responsibilities:
  - Builds statistical and machine learning models
  - Works on predictive and prescriptive analytics
  - Experiments with algorithms to answer complex questions
- They are the **“innovators” who build intelligence on top of data**.

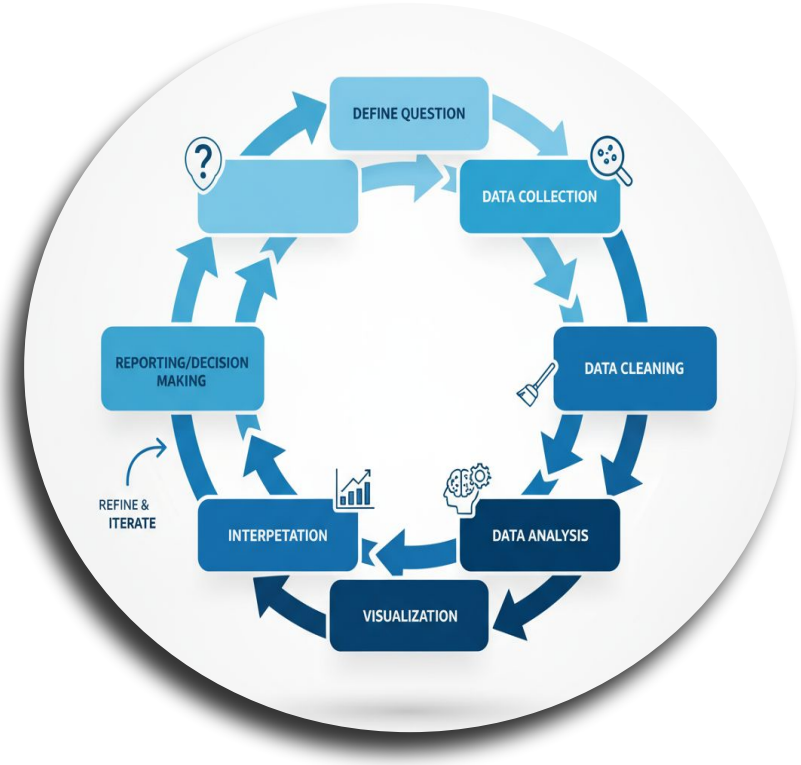


Data Analyst vs Business Analyst: Key Differences

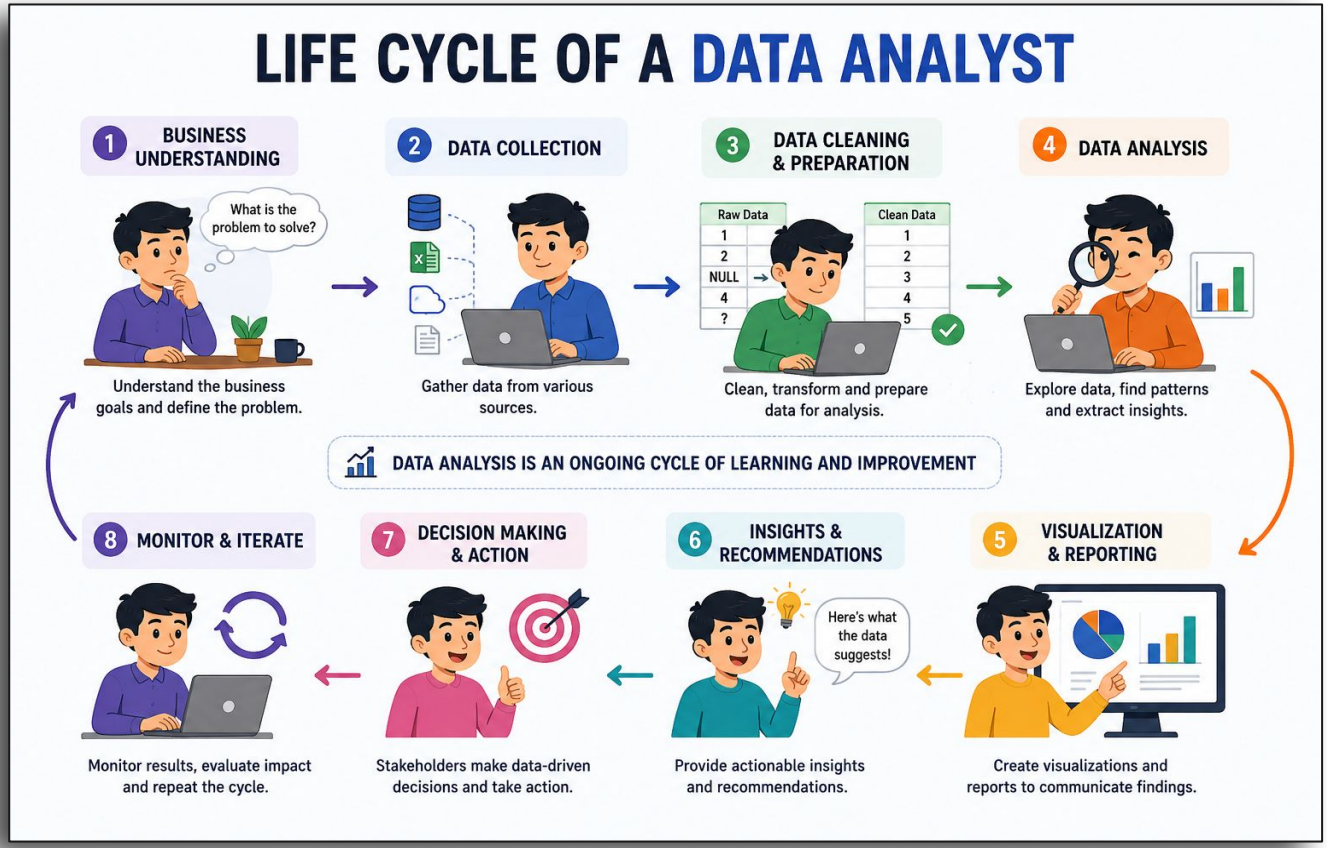
Aspects	Data Analyst	Business Analyst
Focus	Works with data (cleaning, analyzing, visualizing).	Works with business needs (requirements, processes, solutions).
Goal	Extracts insights from data to guide decisions.	Translates business needs into actionable solutions.
Skills	SQL, Python/R, Excel, BI tools, statistics.	Excel, BI tools, Communication, requirement gathering, Reporting
Deliverables	Dashboards, reports, predictive models, data-driven insights.	Requirement documents, workflows, feasibility studies, solution recommendations.
Collaboration	Collaborates more with data engineers, data scientists, and BI developers.	Collaborates more with stakeholders, product managers, and business teams.
Outcome	Answers “What does the data say?”	Answers “What should the business do?”

Data Analyst Lifecycle

Now let see the **Lifecycle of Data Analyst**....



LIFE CYCLE OF A DATA ANALYST



Steps 1: Business Understanding

- Identify the **problem statement** and objectives
- Understand **business goals & KPIs**
- Ask the right **questions to stakeholders**
- Define **scope and success criteria**
- Align analysis with **business needs**

Outcome: Clear direction for data analysis and decision-making

Steps 2: Data Collection

Data Collection

- Gathering raw data from different sources → later cleaned & analyzed.
- Why it matters: Right data = accurate insights; wrong data = misleading results.
- Sources:
    - Databases (SQL/Cloud)
    - Data Warehouse / Data Lake
    - Excel/CSV files
    - APIs (social media, stock prices)
    - Surveys & Feedback
    - IoT/Sensors
    - Public Data Sources

Steps 3 & 4: Data Cleaning & Visualization

Data Cleaning (Preprocessing)

- Ensures dataset is accurate, consistent, ready for analysis.
- Tasks: Remove duplicates, handle missing values, correct errors & data types.
- Why it matters: Dirty data → wrong insights; clean data → trust & efficiency.

Visualization

- Turning data into charts, graphs, dashboards.
- **Tools:** Power BI, Tableau, Excel, Python (Matplotlib/Seaborn).
- Common charts: Bar, Line, Pie, Heatmaps, Dashboards.
- Why it matters: Makes data easy to interpret, highlights patterns/trends.

Steps 5 & 6: Interpretation & Reporting/Decision Making

Interpretation

- Connects insights → real-world causes & actions.
- Approach: Look for patterns → Ask “Why” → Add context → Tell the story.
- Why: Turns charts into actionable strategies.

Reporting & Decision Making

- Reports & dashboards bridge gap between **data & action**.
- Good reporting → faster, confident decisions.
- Data-driven decisions = less guesswork, more success.

Reporting

- Reporting is sharing the analysis in a **clear, structured format**.
- Reports can be **static** (PDF, slides) or **interactive** (Power BI/Tableau dashboards).
- Reports can be in the form of Dashboards, Presentation, Stories, etc.
- Let’s take an example of Report generated in Tableau in the form of Dashboard and story which shows the insights generated from the data.


Story
Layout
New story point
Blank
Duplicate
Sales per Year
Sales per Region
Profitability ...
Categorywise ...
Top 10 ...
Dashboard
Sheet 10
Sheet 11
Dashboard 2
Drag to add text
Show title
Size
Custom size (1516 x 864)
Story Telling Practice
An interactive dashboard combining key visuals—yearly sales trends, regional sales distribution, and category-wise breakdown—offering a holistic view of sales performance for better decision-making.
Visualizing annual sales performance with an area chart, highlighting growth patterns, seasonal fluctuations, and key business trends over the years
A stacked horizontal bar chart illustrating sales performance across different regions, making it easy to compare contributions and identify top-performing areas.
A pie chart showcasing the proportion of sales across different product categories, providing a clear view of the top-performing categories
Sales per Region
Sales
4M
3M
2M
1M
0M
2,822,303
1,600,907
1,248,166
1,100,185
884,423
848,310
806,161
783,773
752,827
725,458
678,781
324,281
66,928
Top 10 Customers
Customer Name
Sales
40K
20K
0K
40,488
37,457
35,551
35,187
35,171
32,310
31,781
30,401
30,288
30,244
Tom Ashbrook
Tamara Chand
Greg Tran
Christopher Conant
Sean Miller
Bart Watters
Natalie Fritzler
Fred Hopkins
Jane Waco
Hunter Lopez
Sales per Year
Sales
4M
2M
0M
2,259,451
2,677,439
3,405,746
4,299,866
2011
2012
2013
2014
Year of Order Date
Profitability across Different Countries
© 2025 Mapbox © OpenStreetMap
Categorywise Sales Distribution
Technology
4,744,557
Furniture
4,110,874
Office Supplies
3,787,070
Data Source
Sales per Year
Sales per Region
Profitability across different cou...
Categorywise Sales distribution
Top 10 Customers
Dashboard
Sheet 10
Sheet 11
Dashboard 2
Story 1

Think and Answer !!!

"We've discussed several tools that are required in Data Analysis . Can you name some of them?"



The Future of Analytics Generative AI

- Analytics is changing fast.
- Analysts need to use AI for faster insights & smarter decisions.
- Now Let’s see How Gen AI helps Analyst



Traditional Workflow vs GenAI-Enhanced Workflow

Workflow Step	Traditional Approach	How GenAI Helps
Data Collection	Collect data from multiple sources.	(No major change – still requires human setup).
Data Cleaning	Fix errors, remove duplicates, format data.	(No major change – but AI tools can sometimes auto-detect errors).
Data Analysis	Manually analyze to find trends & insights.	Quickly **summarizes raw data** into insights.
Visualization	Manually select & create charts/graphs.	**Suggests appropriate charts** to communicate findings better.
Reporting	Write presentations or documents manually.	**Drafts reports automatically**, reducing writing effort.
Main Challenge	Time-consuming & repetitive.	AI speeds up repetitive steps → analyst focuses on decisions.

What is Prompt Engineering?

- A **prompt** = the question or instruction you give to AI.
- Prompt Engineering = **crafting better instructions** so AI gives useful answers.
- Think of AI as a student → the better your question, the better their answer.
- Analysts use prompts to save time in reporting, visualization, and summarization.



Example

Prompt: “Explain sales trend in 2 lines.”

AI Output: “Sales increased in Jan–Feb due to promotions, but fell in March.”



Why Prompt Matters?

- AI does **not read minds** → it follows *exact instructions*.
- Vague instructions = vague output.
- Clear instructions = useful, specific output.

Examples:

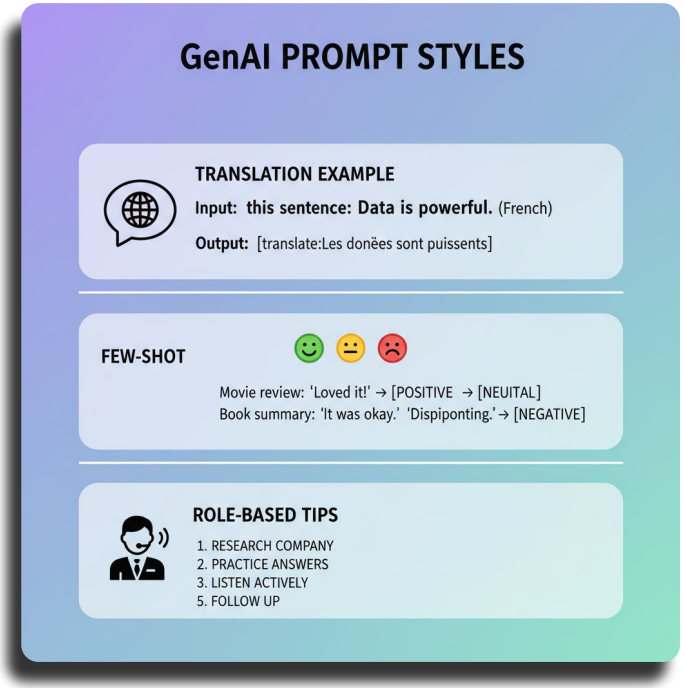
- Bad Prompt: “Tell me about sales.”
  → AI might write 2 pages, unfocused.

- Good Prompt: “Summarize last month’s sales in 3 bullet points for managers.”
  → AI gives short, clear summary.



Simple Types of Prompts

- **Zero-shot Prompting** → Ask without giving examples.
    - Example: “Translate ‘Hello’ into French.”
- **Few-shot Prompting** → Give 1–2 examples first.
    - Example:
        i. Classify review: ‘The food was bad’ → Negative.
        ii. ‘The movie was good’ → Positive.
        iii. Now classify: ‘The service was excellent.’
- **Role-based Prompting** → Tell AI to act like an expert.
    - Example: “You are a nutritionist. Suggest 2 healthy snacks.”



Let’s Discuss

In the analytics workflow, which step do you think usually takes the longest, and why?



✅ Answer : Data Cleaning – The Longest Step

The Challenge

- Data Cleaning & Preparation takes the **longest time** in the workflow.
- **Why?**
  - Raw data has duplicates, missing values, errors, formatting issues.
  - Requires careful checks for accuracy and consistency.

- Analysts spend **60–70% of their time** just on cleaning and preparing data.

How GenAI Helps

- Auto-detects duplicates and anomalies.
- Suggests ways to handle missing values.
- Standardizes formats (dates, text, categories).
- Saves precious time → Analysts focus more on **insights & decisions**.

Let’s See the Magic of AI

*AI is like a **data-cleaning assistant** → fast, smart, and always ready.*

*We’ll explore how tools like **Copilot, Power BI AI, Tableau Pulse, AI-ETL, and AugNLP** make analysts’ jobs easier*

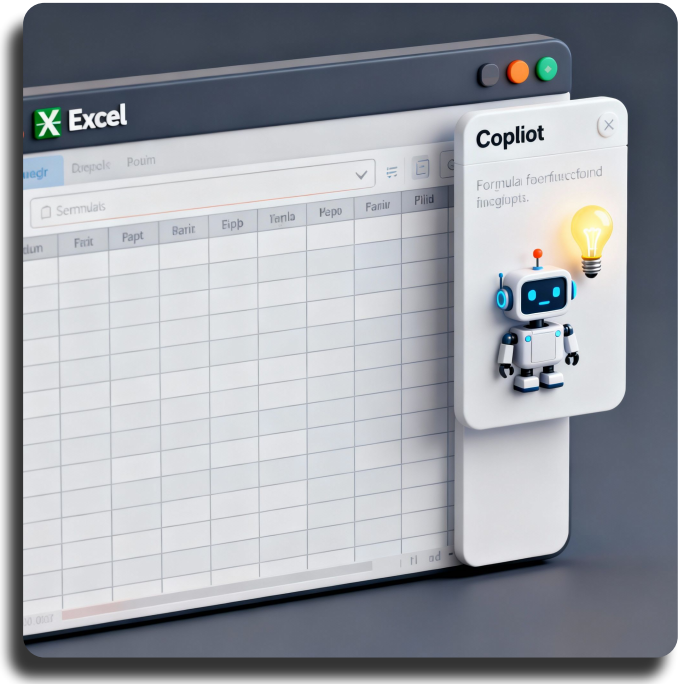


Microsoft Copilot for Excel

How it Helps Analysts:

- Detects **duplicates, blanks, errors** automatically.
- Suggests formulas like **TRIM, LEFT, CONCAT**.
- Explains formulas in **plain English**.

Example:
Ask Copilot → *“Remove extra spaces and split full names into First & Last Name”* → Copilot generates formula.



AI Superpowers in BI – Power BI AI & Tableau AI

We’ve used BI tools to make charts and dashboards. But imagine if your BI tool could **think like an analyst** — finding patterns, highlighting problems, and even sending alerts.

That’s exactly what AI is bringing into **Power BI** and **Tableau**!

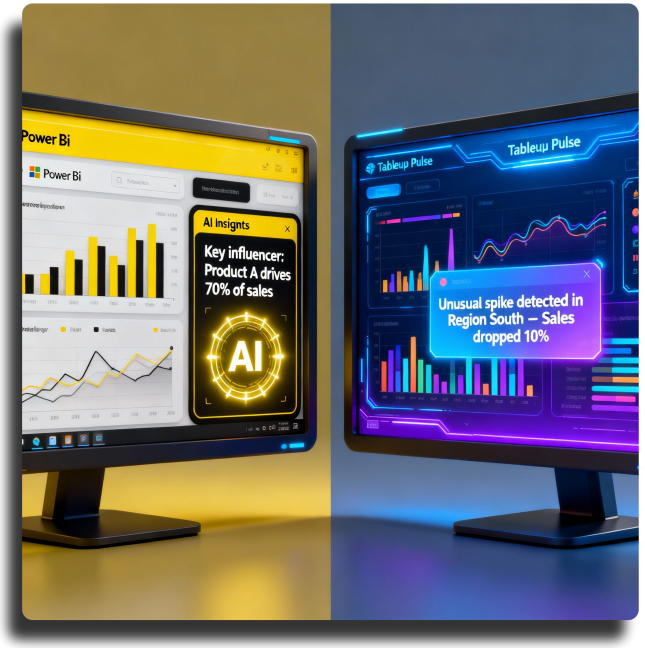
AI features in Power BI

- **Copilot** → AI assistant for chatting and generating reports.
- **AutoML** → Builds machine learning models automatically without coding.
- **Q&A** → Type your question, get instant answers and visuals.
- **AI visuals** → Smart charts that spot trends and make predictions.
- **Decomposition Tree** → Breaks down metrics into contributors
  (e.g., Sales drop → Region → Product).
- **Key Influencers Visual** → Identifies what factors impact KPIs the most.
- **Anomaly Detection** → Spots unusual spikes/drops in trends.



AI features in Tableau

- **Einstein Copilot** → Your AI assistant for personalized data insights.
- **Ask Data & Tableau GPT** → Ask questions in natural language, get visuals.
- **Explain Data** → Understand why data looks unusual.
- **Smart Anomaly Detection** → Highlights odd data points quickly.
- **AI Prep Assistant** → Automates data preparation and cleaning steps.
- **Tableau Pulse** → Real-time monitoring of data,Sends **AI-generated alerts** when patterns look unusual.



AI in ETL – Smart Tools for Smarter Data

*You've seen how AI helps in Excel, BI, and dashboards. But what if AI could also handle the toughest part of analytics — **ETL (Extract, Transform, Load)**?*

Yes! Today's ETL tools are becoming intelligent assistants that clean, transform, and map data automatically.



Popular AI ETL Tools

1️⃣ **Informatica** – *The Data Integrator*

- Uses AI for **smart data integration & automation**
- Learns mapping patterns → reduces manual effort

2️⃣ **Talend** – *The Data Quality Checker*

- Uses ML to **spot duplicates, blanks, errors**
- Suggests **auto-fixes** for cleaner data

3️⃣ **AWS Glue** – *The Serverless Helper*

- AI-powered **serverless ETL**
- Auto-discovers & organizes data in the cloud

4️⃣ **Microsoft Fabric** – *The All-in-One Platform*

- Combines **ETL, storage (OneLake), and analytics**
- AI helps in **dataflows, transformations, and integration with Power BI**

AI Data Augmentation Tools

How They Help (in NLP projects):

- **AugNLP:** Creates extra training data by **paraphrasing & adding synonyms**.
- **TextAttack:** Generates variations of text for **robust ML models**.
- Makes datasets **bigger, richer, and less biased**.

Example:
Original: “Great product!”

AI variations: “Amazing item!”, “Loved this product!”, “Fantastic purchase!”



Human + AI Hybrid Workflow

Explanation: Analysts won't be replaced. Instead, they will work **together with AI**.

- **AI does:**
  - Summaries
  - Draft reports
  - Chart suggestions

- **Analyst does:**
  - Validates correctness
  - Adds business context
  - Makes final decisions



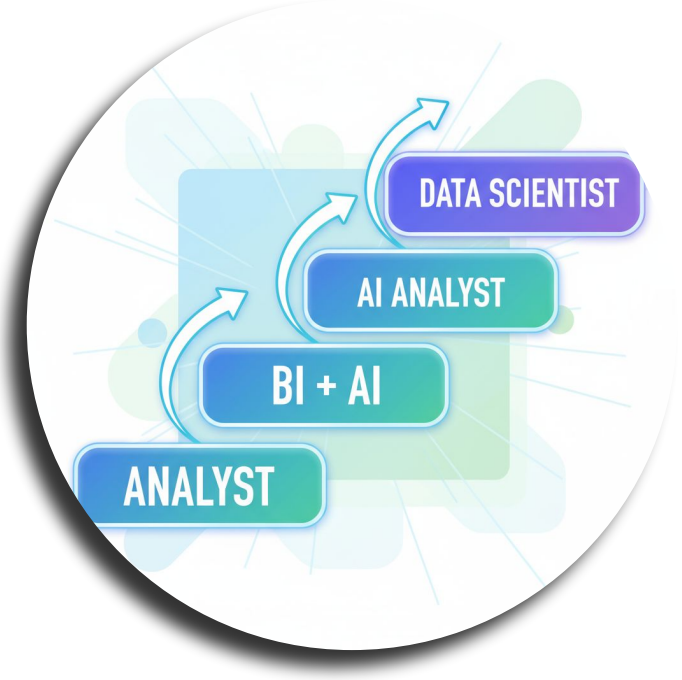
AI is a helper, not a replacement.



Career Pathways in the GenAI Era

Explanation: With AI, analysts can **grow faster** into new roles.

1. **Data Analyst (Entry Level)** – Excel, SQL, dashboards.
2. **BI Analyst with AI** – Uses AI to speed reporting & analysis.
3. **AI Product Analyst** – Works with AI tools/products to improve decision-making.
4. **Data Scientist / AI Specialist** – Builds models, advanced analytics, ML.



Ethics in AI for Analysts

With great tools comes **great responsibility**.

- Always **validate AI outputs** (AI can make mistakes).
- Avoid **bias** in data or prompts.
- Be **transparent** → Don't hide if AI was used.
- Remember → **AI supports, Analyst decides**.

Example: If AI generates a wrong sales forecast, it's the analyst's duty to correct it before presenting.



Applications and Examples

Now Let’s see the applications of Data Analytics Across Industries...



1. Retail

- **Applications:**
  - Recommendation engines analyze purchase history & browsing patterns.
  - Personalized shopping boosts sales.
- **Example:**
  - Amazon: “Customers who bought this also bought…”
  - Flipkart: “Recommended for you” based on past searches.



2. Finance

- **Applications:**
  - Fraud detection using ML models.
  - Credit scoring to assess loan risk.
- **Example:**
  - Banks flag unusual credit card transactions.
  - Loan approvals based on repayment behavior.



3. Healthcare

- **Applications:**
  - Predict outbreaks using hospital, climate, and mobility data.
  - Patient monitoring via wearables (heart rate, glucose).
- **Example:**
  - Fitbit/Apple Watch alerts for irregular heartbeat.
  - COVID-19 dashboards track spread trends.

4. Sports

- **Applications:**
  - Player performance analysis: speed, accuracy, stamina.
  - Strategy planning for team formation & tactics.
- **Example:**
  - Cricket: Analyze batsman weakness vs bowler type.
  - Football: Track player movements to optimize training.

5. Government :

Application

- **Smart Cities**: Managing traffic, electricity, and water supply with real-time data.
- **Policy-Making**: Using data to evaluate programs and plan budgets.

Example

- Traffic signal systems using sensors in Bengaluru/Delhi.
- Data-driven decisions for vaccination campaigns.



Quick Recap Question

“Which industry do you think benefits the most from data analytics, and why?”



Quiz Question

Question:
Which type of analytics answers *“What should we do?*

A. Descriptive Analytics

B. Predictive Analytics

C. Prescriptive Analytics

D. Generative Analytics

Quiz Question

Explanation: *Prescriptive gives action recommendations.*

A. Descriptive Analytics

B. Predictive Analytics

C. Prescriptive Analytics

D. Generative Analytics

Quiz Question

Question:
What is the main goal of data analytics?

A. Store as much data as possible

B. Identify patterns for decision-making

C. Replace human jobs

D. Build databases

Quiz Question

Explanation:

Analytics helps organizations find trends and insights to make better decisions.

A. Store as much data as possible

B. Identify patterns for decision-making

C. Replace human jobs

D. Build databases

Quiz Question

Question:
Why do vague prompts lead to poor results?

A. AI gets confused without clarity

B. AI can’t process them

C. AI ignores vague prompts

D. Vague prompts are too long

Quiz Question

Explanation:
AI works by predicting next from your input. If input is unclear, AI's guess will be random.

A. AI gets confused without clarity

B. AI can't process them

C. AI ignores vague prompts

D. Vague prompts are too long

Recap & Key Takeaways

- **Data Analysts are decision enablers**: Tools are important, but problem-solving mindset and critical thinking matter more.
- **Analytics is evolving**: From descriptive to generative; GenAI automates reporting, saves time, and reduces effort.
- **Hybrid skills are essential**: Combine technical skills (SQL, BI, Python, AI) with human skills (validation, storytelling, critical thinking).
- **AI supports, humans decide**: Clear prompts improve AI results, but human oversight remains crucial.



PW SKILLS

Thank You for Attending the Session !!

This is just the beginning. As we move forward, you'll explore Excel, SQL, Python, BI tools, and Generative AI in more detail. Each session will take you one step closer to becoming a future-ready Data Analyst

Keep Learning, Keep Upskilling!